ARAVINTH K

Senior Embedded Firmware Engineer

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PROFILE SUMMARY

Senior Embedded Firmware Engineer with 5+ years of experience in embedded firmware development, debugging, integration and validation across industrial automation and consumer electronics domains. Skilled in Embedded C, Zephyr RTOS, Embedded Linux and communication protocols including CAN, SPI, I2C, UART and MQTT. Delivered firmware for three commercial Corsair sim-racing products and received several Bosch awards for quality and delivery excellence.

TECHNICAL SKILLS

Programming Languages: Embedded C, C++

RTOS & Operating Systems: Zephyr RTOS, Embedded Linux

Microcontrollers: STM32, TI TIVA C, NXP LPC Series, Atmel AVR/SAM, Microchip PIC

Communication Protocols: CAN, UART, SPI, I2C, RS232, RS485, MQTT

Wireless & Connectivity: Bluetooth, Wi-Fi, GPS, GSM, RFID

Embedded Systems: Firmware Development, Firmware Validation, Embedded Software Design, System Integration, Debugging, Root Cause Analysis

Testing & Quality: Unit Testing (Ceedling), Integration Testing, System Testing, Static Code Analysis (Cppcheck)

Tools & IDEs: Git, Gerrit, Version control, JIRA, GDB, JTAG, Oscilloscope, Logic Analyzer, VS Code, Keil MDK, MPLAB X, Atmel Studio, Code Composer Studio (CCS)

WORK EXPERIENCE

Analyst (Senior Embedded Firmware Engineer) | Bosch Global Software Technologies Pvt. Ltd. *Feb 2023 – Present*

Coimbatore, India

- Developed firmware for 3 commercial Corsair sim-racing products using Zephyr RTOS, including Handbrake, PVGT Porsche and Formula V3 Steering Wheel.
- Resolved 21+ defects across 4 Corsair product lines through defect resolution, root cause analysis, regression testing, as well as feature enhancements, improving product stability and reliability.
- Led and mentored a team of 2 junior firmware engineers, providing technical guidance, task planning, code reviews, and project onboarding support.

Embedded Software Developer | Kalycito *Apr 2022 – Jan 2023*

Coimbatore, India

- Supported C++ application software on Linux for EdgeXConnect, a configurable no-code OPC UA Server platform for industrial IoT environments.
- Performed defect fixes, OPC UA node modifications, and release validation activities in an Agile development environment.
- Improved code organization and maintainability by restructuring the project using Git submodules and contributing to release validation activities.

Embedded Software Developer | Petrus Technologies Pvt. Ltd. *Jun 2021 – Apr 2022*

Coimbatore, India

- Sole firmware engineer for a 3-Layer Conveyor System (textile automation) delivered firmware development, system integration, testing and on-site deployment end-to-end in under 4 months.
- Calibrated and fine-tuned stepper motor control firmware for an Automatic Cone Winding Machine, implementing bug fixes and enhancements to improve automation performance and system reliability.
- Designed and validated a pulse oximeter proof-of-concept using the MAX30100 sensor, implementing SpO_2 and heart-rate measurement algorithms with ±2% accuracy.

KEY PROJECTS

Corsair Sustainability, PVGT Porsche & Formula V3 Steering Wheels *Bosch · 2024 – Present*

Stack: *Embedded C, Zephyr RTOS, NXP LPC, CAN, SPI, I2C*

- Delivered sustainability support for multiple Corsair (Endor) sim-racing products through defect resolution, root cause analysis, regression testing and firmware enhancements.
- Developed and customized firmware for the PVGT Porsche and Formula V3 Steering Wheels using Zephyr RTOS by extending an existing firmware framework. Implemented product-specific features, performed product integration, system-level testing and supported release activities.

Corsair Handbrake — Sim Racing Peripheral Firmware *Bosch · 2023 - 2024*

Stack: *Embedded C, Zephyr RTOS, NXP LPC, CAN Bus, Sensor Interface, EEPROM*

- Developed Zephyr RTOS firmware for a commercial sim-racing handbrake, implementing HX717 sensor acquisition, CAN communication, EEPROM-based profile storage and bootloader integration.
- Coordinated product release activities, including wheelbase compatibility validation, regression testing, and launch sign-off.

EdgeXConnect — Industrial No-Code OPC UA Server *Kalycito · 2022 – 2023*

Stack: *C++, Linux, OPC UA SDK, CMake*

- Maintained and enhanced a C++ OPC UA server application on Linux, resolving defects and contributing to integration and system testing.

AWARDS & RECOGNITION

Best Tyro — Bosch

Recognized for quick adaptation, strong commitment, and high productivity within a short ramp-up period.

Catalyst — Bosch

Excellent defect identification in Corsair Steering Wheel firmware ahead of the product launch schedule.

Growth Driver — Bosch

Outstanding contribution to the Corsair firmware sustainability initiative.

EDUCATION

B.E — Electronics and Communication Engineering *2013 – 2017*

Nandha College of Technology, Erode